Git Hub and GIT

What you'll learn (end-to-end)

1. What Git & GitHub are (very briefly)
2. Prerequisites
3. Sign in to GitHub from VS Code
4. Create a project in VS Code
5. Initialize Git repository
6. Write example code (my own)
7. Commit changes
8. Create a GitHub repository
9. Push code to GitHub
10. Make changes & push again
11. Pull changes
12. Common Git commands you should remember
13. Typical workflow (real life)

1️⃣ Git vs GitHub (quick clarity)

- **Git** → Version control tool (tracks changes locally on your machine)
- **GitHub** → Cloud platform to host Git repositories and collaborate

VS Code acts as a **front-end** to Git and GitHub.

Prerequisites (one-time setup)

Make sure you have:

✅ Install Git

Check:

```
git --version
```

``

If not installed:

- Download from: https://git-scm.com

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

Microsoft Windows [Version 10.0.22631.6783]
(c) Microsoft Corporation. All rights reserved.

C:\Users\YZW0KHP\OneDrive - ups\personal documents\self learning projects\git hub vs code demo>git --version
git version 2.53.0.windows.2

C:\Users\YZW0KHP\OneDrive - ups\personal documents\self learning projects\git hub vs code demo>
```

✅ Install VS Code

- Download: https://code.visualstudio.com

✅ Create a GitHub account

- https://github.com

3️⃣ Sign in to GitHub from VS Code

1. Open **VS Code**
2. Click **Accounts (** 👤 **)** in bottom-left
3. Click **Sign in with GitHub**
4. Browser opens → Authorize VS Code
5. Done ✅

👉 VS Code now has permission to push/pull code.

4️⃣ Create an example project in VS Code

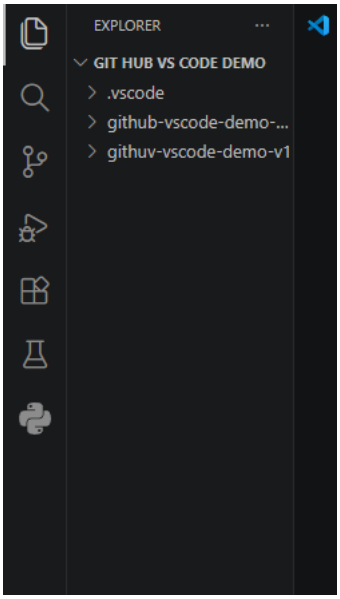
Step 1: Create a folder

mkdir github-vscode-demo

cd github-vscode-demo

```
C:\Users\YZW0KHP\OneDrive - ups\personal documents\self learning projects\git hub vs code demo>mkdir github-vscode-demo-v1

C:\Users\YZW0KHP\OneDrive - ups\personal documents\self learning projects\git hub vs code demo>
```



Change of directory

```
C:\Users\YZW0KHP\OneDrive - ups\personal documents\self learning projects\git hub vs code demo>cd github-vscode-demo-v1

C:\Users\YZW0KHP\OneDrive - ups\personal documents\self learning projects\git hub vs code demo\github-vscode-demo-v1>
```

Step 2: Open the folder in VS Code

code github-vscode-demo

```
C:\Users\YZW0KHP\OneDrive - ups\personal documents\self learning projects\git hub vs code demo\github-vscode-demo-v1>code github-vscode-demo-v1

C:\Users\YZW0KHP\OneDrive - ups\personal documents\self learning projects\git hub vs code demo\github-vscode-demo-v1>
```

5️⃣ Initialize Git repository

Option A: Using VS Code UI

1. Open **Source Control** tab (Ctrl + Shift + G)
2. Click **Initialize Repository**

Option B: Using terminal

git init

``

✅ Your folder is now a Git repository.

```
C:\Users\YZW0KHP\OneDrive - ups\personal documents\self learning projects\git hub vs code demo\github-vscode-demo-v1>git init
Initialized empty Git repository in C:/Users/YZW0KHP/OneDrive - ups/personal documents/self learning projects/git hub vs code demo/github-vscode-demo-v1
/.git/
```

6️⃣ Create example code (our own)

Create file: **app.py**

```
def greet(name):
    return f"Hello, {name}! Welcome to GitHub with VS Code."


if __name__ == "__main__":
    user = input("Enter your name: ")
    print(greet(user))
```

Create file: **README.md**

```
# GitHub + VS Code Demo


This is a simple Python project to demonstrate:
- Git
- GitHub
- VS Code integration


## How to run
```bash
python app.py
```
```

7️⃣ Stage & Commit changes

🔹 Using VS Code UI

1. Open **Source Control**

2. You'll see `app.py` and `README.md`

3. Click ** ➕ (Stage All Changes)**

4. Enter commit message:

Initial commit: add greeting app

6. Click **✓ Commit**

--- ### 🔹 Using Git commands (what VS Code runs internally)

```bash

git status

```

```
C:\Users\YZW0KHP\OneDrive - ups\personal documents\self learning projects\git hub vs code demo\github-vscode-demo-v1>git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   README.md
        new file:   app.py
        new file:   github-vscode-demo-v1
```

git add

```
C:\Users\YZW0KHP\OneDrive - ups\personal documents\self learning projects\git hub vs code demo\github-vscode-demo-v1>git add .
```

git commit -m "Initial commit: add greeting app"

```
C:\Users\YZW0KHP\OneDrive - ups\personal documents\self learning projects\git hub vs code demo\github-vscode-demo-v1>git commit -m "Initial commit: add
greeting app"
[master (root-commit) c7ac360] Initial commit: add greeting app
 Committer: Titus Sam (YZW0KHP) <YZW0KHP@aad.ups.com>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly:

    git config --global user.name "Your Name"
    git config --global user.email you@example.com

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

 3 files changed, 17 insertions(+)
 create mode 100644 README.md
 create mode 100644 app.py
 create mode 100644 github-vscode-demo-v1
```

8️⃣ Create GitHub repository & push code

Option A (Easiest): From VS Code

1. Source Control → **Publish Branch**
2. Choose:
    - Public or Private
3. VS Code automatically:
    - Creates repo on GitHub
    - Adds remote
    - Pushes code

✅ Done 🎉

Option B: Manually using commands

Step 1: Create empty repo on GitHub

- Name: github-vscode-demo
- Don't add README

Step 2: Link local repo to GitHub

git remote add origin https://github.com/<your-username>/github-vscode-demo.git

Step 3: Push code

git branch -M main

git push -u origin main

``

9️⃣ **Make a change & push again (real workflow)**

Edit **app.py**

def greet(name):

return f"Hello, {name}! Welcome to GitHub + VS Code 🚀"

Commit & push

VS Code UI

1. Stage changes
2. Commit:

Update greeting message
*3. Click **Sync Changes***
OR Git commands

git add app.py

git commit -m "Update greeting message"

git push

```

```

🔟 Pull changes from GitHub

If someone else changed code OR you edited on GitHub website:

git pull

```

```

Or in VS Code:

- ***Source Control → Sync Changes***

1️⃣1️⃣ Common Git commands you must know

git status # Check current state

git add . # Stage all changes

git commit -m "" # Save a version

git push # Upload to GitHub

git pull # Download updates

git log # Commit history

git branch # List branches

git checkout -b feature-x # Create new branch

1️⃣ 2️⃣ Typical real-world workflow (daily)

1. git pull

2. Write code in VS Code

3. git add .

4. git commit -m "Meaningful message"

5. git push

```

✅ ***What VS Code is doing for you***

| ***VS Code Feature*** | ***Git Command*** |
|---|---|
| *Initialize repo* | *git init* |
| *Stage changes* | *git add* |
| *Commit* | *git commit* |
| *Publish Branch* | *git push* |
| *Sync Changes* | *git pull + push* |

***Summary***

*You now know:*

- *How GitHub works with VS Code*
- *How to create & push real code*
- *Both **UI + command-line***
- *A complete **end-to-end workflow***

From Gemini Version of notes
```

To use GitHub effectively, you need **Git** installed on your computer. While the **Command Line Interface (CLI)** gives you full control through text, **VS Code** offers a user-friendly graphical interface to do the same tasks faster.

🛠️ Prerequisites

Before starting, ensure you have the following:

- **GitHub Account:** Sign up at GitHub.com.
- **Git Installed:** Download and install from git-scm.com.
- **Identify Yourself:** Open your terminal and set your identity so GitHub knows who is making changes:

Bash

git config --global user.name "Your Name"

git config --global user.email your_email@example.com

💻 Method 1: Using Command Line (Git CLI)

The command line is powerful because it works in any environment

1. Initializing or Cloning

- **New project:** Navigate to your folder and type git init to start tracking it.
- **Existing project:** Copy the URL from GitHub and type git clone <URL> to download it.

2. The Standard Workflow

- Follow these four commands to save and share your work

1. **Check status:** git status (See what has changed).
2. **Stage changes:** git add . (Prepare all changed files for a save).
3. **Commit:** git commit -m "Your message" (Save a snapshot locally).
4. **Push:** git push origin main (Upload your save to GitHub).

Method 2: Using Visual Studio Code (GUI)

VS Code has Git support built-in, making it much more visual.

1. Connect to GitHub

- Click the **Accounts** icon (bottom left) and select **Sign in to Sync Settings** or perform any Git action to trigger a sign-in prompt.
- Authenticate in your browser when prompted

2. Clone a Repository

1. Press Ctrl + Shift + P (Windows) or Cmd + Shift + P (Mac) to open the **Command Palette**.
2. Type **"Git: Clone"** and select it.
3. Choose **Clone from GitHub** and pick your repository from the list

3. The Source Control Tab

Open the **Source Control** icon (the branch symbol on the left sidebar)

- **Stage:** Click the **+** icon next to changed files.
- **Commit:** Type your message in the text box and click the **Commit** checkmark.
- **Push/Pull:** Click the **Sync Changes** button or the "three dots" menu for more options.

🔄 **Quick Comparison**

Feature	Command Line (CLI)	VS Code (GUI)
Speed	Faster for experts	Faster for beginners
Visibility	Text-based only	Visual "Diff" (side-by-side)
Capability	Access to 100% of Git	Access to most common tasks

Best For	Complex automation/scripts	Everyday coding and reviewing